

COMP5329 26s1 Final Practice

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W1 Introduction, Evaluation, and Deep Learning Problem Framing

Q1. Calculation Semester 1 2020 Final (modified)

A classifier outputs probabilities $p = (1/8, 1/2, 1/4, 1/8)$ for four classes. The target distribution is $y = (0, 1/2, 1/4, 1/4)$. Compute the cross-entropy $-\sum_i y_i \log_2 p_i$ and explain what would happen if probability mass assigned to a target-supported class were zero.

Q2. MCQ

Which statement best describes one epoch in deep learning?

- (A) One optimizer update.
- (B) One complete pass through the training set.
- (C) One forward pass of one sample.
- (D) The period after validation accuracy peaks.

Q3. True/False

True or false: if a model has 98% training accuracy and 98% test accuracy, it must be well calibrated. Justify briefly.

Q4. Short Answer

A hospital wants to classify X-ray images as normal or suspicious. State the input, target, model output, suitable loss, and two evaluation metrics. Explain why accuracy alone could be misleading.

Q5. Calculation

A binary classifier produces TP=42, FP=18, FN=8, TN=132. Compute precision, recall, and F1. Which metric should be emphasized if missing positives is costly?

Q6. Long Question

A team reports only training accuracy for a deep model trained on an imbalanced dataset. Diagnose two methodological problems and propose a stronger evaluation protocol.

Q7. MCQ

For a single-label 5-class image classifier, which output/loss pairing is most appropriate?

- (A) Five sigmoid outputs with MSE.
- (B) Five logits with softmax cross-entropy.
- (C) One scalar output with MSE.
- (D) Five probabilities with no loss.

Q8. Table/Diagram

Complete the table with one main use and one trap for accuracy, precision, recall, and F1.

Q9. Short Answer

A softmax model assigns 0.99 probability to wrong class predictions on several out-of-distribution inputs. Explain why high confidence does not imply correctness and name one mitigation.

Q10. Long Question

Design an ablation study for a new augmentation method. Specify baseline, controlled variables, outcome metrics, and what conclusion would be invalid without further evidence.



W2 Multilayer Neural Networks and Optimisation

Q1. Calculation Semester 1 2021 Final (modified)

A fully connected network has 10 inputs, one hidden layer with H units, and 10 outputs. There are no biases. A minibatch of 50 cases and produces 1,000,000 per-case gradient calculations $\partial L / \partial w$. Find the number of training cases N , the number of minibatch optimizer steps, and the total scalar weight-update operations.

Q2. Short Answer Semester 1 2024 Final (modified)

For a hidden unit $y_2 = g(w_5x_1 + w_6x_2)$ and output $z = w_1y_1 + w_2y_2$, with $L = (z - t)^2$, derive $\partial L / \partial w_5$.

Q3. MCQ

Which activation is most likely to suffer from saturated near-zero gradients for large positive or negative inputs?

- (A) ReLU
- (B) Sigmoid
- (C) Identity
- (D) Max pooling

Q4. Calculation

Let $X \in \mathbb{R}^{32 \times 20}$ and $W \in \mathbb{R}^{20 \times 5}$, with logits $Z = XW + b$ and loss gradient $\Delta = \partial L / \partial Z \in \mathbb{R}^{32 \times 5}$. State the shape and formula for $\partial L / \partial W$.

Q5. Short Answer

A student increases the learning rate by 500% after observing poor validation accuracy. Give two reasons this may fail and two safer debugging steps.

Q6. True/False

True or false: Adam always generalizes better than SGD with momentum because it adapts the learning rate per parameter.

Q7. Calculation

Compute softmax probabilities for logits $(2, 1, 1)$ in exact exponential form. What is the gradient of CE loss wrt logits if the true class is the first class?

Q8. Long Question

Explain why residual connections help train very deep networks. Your answer should mention gradient flow and the representation learned by a residual block.

Q9. Short Answer

Compare Xavier and He initialization. Which activation family motivates each?

Q10. Long Question

A model's loss is NaN after two updates. Provide a prioritized debugging checklist.



W3 Regularisation in Deep Learning

Q1. MCQ Semester 1 2021 Final (modified)

Which statement about dropout is most accurate?

- (A) It is normally used instead of all regularisation.
- (B) It creates many separate full copies of a network at training time.
- (C) It randomly masks activations and can be interpreted as approximate model averaging.
- (D) It uses only a random fraction of the training set.

Q2. Short Answer Semester 1 2024 Final (modified)

Give four strategies to mitigate overfitting in deep learning. For each, state one advantage and one limitation.

Q3. True/False

True or false: batch normalization and dropout are identical because both randomly change hidden activations during training.

Q4. Calculation

For a scalar weight $w = 3$, data gradient $\partial L / \partial w = 0.4$, learning rate $\eta = 0.1$, and L_2 penalty λw^2 with $\lambda = 0.05$, compute one gradient descent update.

Q5. Short Answer

Why can data augmentation improve generalisation, and when can it hurt?

Q6. MCQ

Which statement about batch normalization is correct?

- (A) It has trainable γ, β parameters.
- (B) It always removes the need for validation.
- (C) It must only be used after fully connected layers.
- (D) It makes model capacity zero.

Q7. Long Question

A validation curve improves for 8 epochs then worsens while training loss keeps falling. Explain the likely diagnosis and design an early-stopping rule.

Q8. Short Answer

Compare L_1 and L_2 regularisation in terms of sparsity and weight shrinkage.

Q9. Table/Diagram

Fill a two-column table mapping each failure to a likely fix: tiny batch BN instability; overconfident predictions; memorisation of training images; exploding gradients.

Q10. Long Question

A team tunes dropout rate using the test set because the validation set is small. Explain the problem and propose an alternative.



W4 Convolutional Neural Networks

Q1. Calculation Semester 1 2020 Final (modified)

Input has 3 channels, each 15×15 . A convolution layer has 5 output feature maps, 3×3 kernels, stride 2, and no padding. How many output neurons are there?

Q2. Short Answer Semester 1 2024 Final (modified)

Explain two fundamental differences between convolutional layers in CNNs and self-attention layers in Vision Transformers.

Q3. Calculation

An RGB image 224×224 passes through a conv layer with $C_{out} = 64$, kernel 7×7 , stride 2, padding 3. Compute output spatial size and parameter count including biases.

Q4. MCQ

Why are stacked 3×3 convolutions often preferred to one very large convolution?

- (A) They cannot learn nonlinear features.
- (B) They increase receptive field with fewer parameters and more nonlinearities.
- (C) They remove the need for training data.
- (D) They make all outputs invariant to class labels.

Q5. Short Answer

What does a residual skip connection solve, and when is a projection shortcut needed?

Q6. True/False

True or false: a 21×21 CNN filter exactly mimics transformer attention because both see a large region.

Q7. Table/Diagram

Compare VGG, Inception, ResNet, and U-Net by core idea and typical use.

Q8. Long Question

A CNN has high accuracy on centered objects but fails when objects shift to image corners. Diagnose possible causes and fixes.

Q9. Short Answer

Define IoU and explain its role in object detection evaluation.

Q10. Long Question

You must implement a VGG-like block in PyTorch. List the layers and forward-pass shape reasoning for input $(B, 3, 224, 224)$ through two same-padded 3×3 convs to 64 channels then 2×2 max pool.



W5 Graph Neural Networks

Q1. Long Question Semester 1 2024 Final (modified)

Compare GCNs and CNNs for image recognition. In what image settings might a GCN be more suitable than a CNN?

Q2. Calculation

Given an undirected graph with adjacency A and node features $H \in \mathbb{R}^{n \times d}$, state the normalized GCN update using self-loops and give the shape of the output if $W \in \mathbb{R}^{d \times k}$.

Q3. MCQ

What property should a graph-level readout function usually have?

- (A) Depend on node ordering.
- (B) Be permutation invariant.
- (C) Delete all edge information before aggregation.
- (D) Require a square feature matrix.

Q4. Short Answer

Explain oversmoothing in deep GNNs and give two mitigation strategies.

Q5. True/False

True or false: adding self-loops in a GCN is unnecessary because neighbours always contain complete information about a node.

Q6. Long Question

A graph classifier performs well on training graphs but fails on new graphs with different degree distributions. Diagnose the issue and propose robust evaluation.

Q7. Table/Diagram

Compare GCN, GraphSAGE, GAT, and GIN by aggregation idea and one strength.

Q8. Short Answer

Why can a poor graph construction make a strong GNN fail?

Q9. Calculation

A sparse graph has $|E| = 50,000$, hidden size $d = 64$, and one message-passing layer roughly costs $O(|E|d)$. Estimate the number of edge-feature multiplications up to a constant.

Q10. Long Question

Design a GNN for skeleton-based action recognition. Specify nodes, edges, input features, model output, and one failure mode.



W6 Consolidation: Gradients, Shapes, and Exam Reasoning

Q1. Calculation Semester 1 2020 Final (modified)

A fully connected network has 9 inputs, 5 outputs, one tanh hidden layer with biases only in the hidden layer, and parameters. Find the number of hidden units.

Q2. Short Answer

Give a compact checklist for deriving a backpropagation gradient by hand.

Q3. Calculation

Three matrices have shapes $A : m \times n$, $B : n \times p$, $C : p \times q$. Give the scalar multiplication cost for $(AB)C$ and $A(BC)$.

Q4. MCQ

In a PyTorch training loop, which order is standard?

- (A) backward, loss, forward, step.
- (B) zero gradients, forward, loss, backward, optimizer step.
- (C) optimizer step, zero gradients, validation, loss.
- (D) loss, zero gradients, save, no backward.

Q5. True/False

True or false: calling 'model.eval()' changes the learned weights of a network.

Q6. Long Question

A student's answer to an architecture comparison only lists definitions. Rewrite the answer structure so it can earn marks for reasoning.

Q7. Calculation

A model trained for 12 epochs has 800 training samples and batch size 40. How many optimizer updates occur if every batch is used once per epoch?

Q8. Short Answer

Explain why overfitting a tiny batch is a useful debugging test.

Q9. Table/Diagram

Make a four-row troubleshooting table for: no learning; train good val bad; train and val both bad; val metric noisy.

Q10. Long Question

An exam asks for advantages and disadvantages of a defence method. What minimum components should your answer include?



W7 RNNs, LSTMs, and Transformers

Q1. Long Question Semester 1 2021 Final (modified)

For a simple RNN unrolled over time, explain why backpropagation through time can suffer from vanishing or exploding gradients.

Q2. Short Answer Semester 1 2023 Final (modified)

In PyTorch RNN training, why may a hidden state need to be detached between batches? Explain the difference between `detach()` and `detach_()` in intent.

Q3. Calculation

An RNN has input dimension 30, hidden dimension 50, and output dimension 10. Count trainable parameters for input-hidden weights, hidden-hidden weights, hidden bias, hidden-output weights, and output bias.

Q4. MCQ

Why is layer normalization often preferred to batch normalization in transformers?

- (A) It normalizes across a token's hidden features and is stable for variable batch/sequence settings.
- (B) It requires future tokens.
- (C) It deletes residual connections.
- (D) It only works for images.

Q5. Short Answer

Write the scaled dot-product attention formula and explain the role of $\sqrt{d_k}$.

Q6. True/False

True or false: positional encoding is unnecessary in a vanilla transformer encoder because self-attention already knows word order.

Q7. Long Question

Compare LSTM and transformer models for long sequences under training-time parallelism, memory, and inductive bias.

Q8. Calculation

For sequence length $T = 1024$ and hidden attention dimension ignored, compare the number of pairwise attention scores in full self-attention vs a local window of 32 tokens per token.

Q9. Short Answer

Explain exposure bias in teacher forcing and name one mitigation.

Q10. Table/Diagram

Compare RNN, GRU, LSTM, and transformer in one table by memory mechanism and main limitation.



W8 Beyond Transformer Architectures

Q1. Short Answer

What practical problem motivates Mamba, RWKV, and other beyond-transformer sequence models?

Q2. MCQ

Which claim is most defensible?

- (A) Subquadratic sequence models are always more accurate than attention.
- (B) State-space models compress history into state and can be efficient for long sequences.
- (C) KAN removes the need for data.
- (D) KV cache makes attention training $O(1)$.

Q3. Long Question

Compare full attention and a selective state-space model for a streaming sensor task. Address memory, latency, and failure modes.

Q4. True/False

True or false: if a model has linear-time inference, it necessarily has better factual recall over a million-token context than full attention.

Q5. Short Answer

Explain the purpose of a KV cache in autoregressive generation and one limitation.

Q6. Table/Diagram

Make a compact comparison of attention, Mamba/SSM, RWKV, and KAN as beyond-standard-transformer design ideas by core idea and risk.

Q7. Calculation

A full attention layer stores a $T \times T$ attention matrix. If T doubles from 2048 to 4096, by what factor does this matrix size increase?

Q8. Long Question

A paper claims its new long-context model is superior because it handles 128k tokens. What controls and tests would you require?

Q9. Short Answer

What is the difference between exact pairwise token interaction and compressed-state sequence modelling?

Q10. Long Question

Argue for a hybrid attention-SSM architecture in a long document QA system.



W9 Multimodal Foundation Models

Q1. Long Question Semester 1 2023 Final (modified)

A user wants a foundation model to help with scheduling, communication, information retrieval, and travel planning tasks, then write what information a good prompt should include and explain why specificity matters.

Q2. Short Answer

Define a vision-language model and explain the role of cross-attention.

Q3. Calculation

An image is split into 16×16 patches. For a 224×224 image, how many patch tokens are produced before adding special tokens?

Q4. MCQ

CLIP-style contrastive training primarily tries to:

- (A) reconstruct pixels exactly.
- (B) align matched image-text embeddings and separate mismatches.
- (C) predict the next video frame only.
- (D) remove all text tokens.

Q5. True/False

True or false: a multimodal model cannot hallucinate because it has access to images.

Q6. Long Question

Design an evaluation protocol for a VLM used in medical image report assistance.

Q7. Table/Diagram

Compare early fusion, late fusion, and interleaved fusion for multimodal modelling.

Q8. Short Answer

What is a modality gap, and how can contrastive pretraining reduce it?

Q9. Long Question

A VLM answers a counting question incorrectly. Give three possible causes and three fixes or tests.

Q10. Short Answer

Explain why video-language models are more expensive than image-language models.



W10 Deep Reinforcement Learning

Q1. Short Answer

Define an MDP and name the role of S, A, P, R, γ .

Q2. Calculation

If rewards are $r_0 = 1, r_1 = 2, r_2 = 4$ and $\gamma = 0.5$, compute the discounted return from time 0 for these three rewards.

Q3. MCQ

What is the main purpose of a target network in DQN?

- (A) Provide a slower-moving bootstrap target.
- (B) Increase image resolution.
- (C) Delete replay memory.
- (D) Make rewards supervised labels.

Q4. True/False

True or false: RL differs from supervised learning because actions can affect future data observed by the agent.

Q5. Short Answer

Explain the exploration-exploitation trade-off and give one exploration strategy.

Q6. Long Question

A reward function for a cleaning robot gives reward only for speed. Predict a failure mode and redesign the reward.

Q7. Calculation

For Q-learning target $y = r + \gamma \max_{a'} Q(s', a')$, let $r = 2, \gamma = 0.9$, and next-state Q-values $(1, 3, 0)$. Compute y .

Q8. Table/Diagram

Compare DQN, policy gradient, actor-critic, and DPO/RLHF-style preference optimisation by objective signal and one limitation.

Q9. Long Question

Explain how an advantage baseline reduces variance in policy gradient without changing the expected gradient direction.

Q10. Short Answer

What is off-policy learning, and why can it be both useful and risky?



W11 Self-Supervised Representation Learning

Q1. Short Answer Semester 1 2020 Final (modified)

Contractive autoencoders penalize sensitivity of hidden representations to inputs. Explain the intuition of a penalty $\lambda \sum_i \|\nabla_x h_i\|^2$.

Q2. MCQ

Which is a self-supervised pretext task?

- (A) Mask patches and reconstruct them.
- (B) Use human labels only.
- (C) Report test labels during training.
- (D) Tune on the test set.

Q3. Short Answer

Compare linear probing and full fine-tuning as representation evaluations.

Q4. Calculation

In SimCLR, a batch contains 128 images and two augmented views per image. For one view, how many positive views and how many other views are potential negatives, ignoring same-image duplicate only?

Q5. True/False

True or false: contrastive learning is unaffected by false negatives because all non-matching pairs are truly different classes.

Q6. Long Question

Explain why MAE masks a high proportion of image patches and why this can be efficient for ViT pretraining.

Q7. Table/Diagram

Compare autoencoder, contrastive learning, MAE, and CLIP by training signal and failure mode.

Q8. Short Answer

What is representation collapse and name two mechanisms to avoid it.

Q9. Long Question

A company has millions of unlabeled product images and 2,000 labeled defect examples. Propose an SSL pipeline.

Q10. Short Answer

Why must augmentation choice be domain-aware in self-supervised learning?



W12 Deep Generative Models

Q1. MCQ Semester 1 2024 Final (modified)

In an ideal GAN equilibrium where generated and real data distributions match, what should the discriminator output probability?

- (A) 0
- (B) 0.5
- (C) 1
- (D) ∞

Q2. Long Question Semester 1 2024 Final (modified)

A text-to-video system uses diffusion rather than GANs. Give two reasons diffusion may be preferred for text-to-video and two remaining limitations.

Q3. Short Answer

Write the GAN minimax objective and explain the roles of G and D .

Q4. Calculation

In a diffusion forward process $x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$. If $\bar{\alpha}_t = 0.81$, what are the coefficients on x_0 and ϵ ?

Q5. MCQ

Which model explicitly uses an encoder distribution and KL regularisation against a prior?

- (A) VAE
- (B) Standard GAN
- (C) Max pooling
- (D) DQN

Q6. True/False

True or false: mode collapse is a typical concern in GAN training.

Q7. Table/Diagram

Compare GAN, VAE, diffusion, and flow matching by strength and weakness.

Q8. Long Question

Design an audio extension for a video generation model. Mention data, architecture, alignment, and evaluation.

Q9. Short Answer

What is classifier-free guidance and what trade-off does increasing guidance weight create?

Q10. Long Question

An adversarially perturbed image is visually unchanged but confidently misclassified. Explain the mechanism and two defences.



Answer Key

W1 Introduction, Evaluation, and Deep Learning Problem Framing

- Q1. $-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{4} \log_2 \frac{1}{4} - \frac{1}{4} \log_2 \frac{1}{8} = \frac{1}{2} + \frac{1}{2} + \frac{3}{4} = 1.75$. If $p_i = 0$ for a class with $y_i > 0$, the loss is infinity due to smoothing/clipping.
- Q2. B. One epoch is one complete pass through the training set; it may contain many minibatch updates.
- Q3. False. Accuracy and calibration are different; a model can be accurate but systematically over- or under-confident.
- Q4. Input image tensor; target binary label; output one logit/probability; binary cross-entropy; metrics such as recall, precision, AUROC, calibration. Accuracy can hide false negatives or class imbalance.
- Q5. Precision = $42/(42 + 18) = 0.70$; recall = $42/(42 + 8) = 0.84$; F1 = $2(0.70)(0.84)/(1.54) \approx 0.764$. Emphasize recall when FN are costly.
- Q6. Problems: training accuracy measures memorization and imbalance can inflate accuracy. Use stratified split before pre-processing, validation for model selection, held-out test once, class-sensitive metrics, calibration, confidence intervals, and error analysis.
- Q7. B. Multiclass single-label classification normally uses logits and softmax cross-entropy.
- Q8. Accuracy: overall correctness, trap imbalance. Precision: avoid FP, trap ignores FN. Recall: avoid FN, trap ignores FP. F1: balances precision/recall, trap hides threshold/calibration.
- Q9. Softmax is normalized relative logit score, not epistemic certainty; OOD inputs can lie far from training distribution. Mitigate with calibration, OOD detection, abstention, ensembles, or robust training.
- Q10. Compare same architecture/training budget with and without augmentation; hold data split, seeds, LR, epochs fixed; report validation/test metrics and variance. Invalid: claiming general superiority from one dataset/seed or from changed training budget.

W2 Multilayer Neural Networks and Optimisation

- Q1. Per-case gradient calculations equal weights $\times 2500$, so weights = 400. Since $10H + 10H = 20H = 400$, $H = 20$. There are 50 minibatch optimizer steps. Each step updates 400 scalar weights, so there are $50 \times 400 = 20,000$ scalar weight-update operations.
- Q2. $\partial L / \partial w_5 = 2(z - t) w_2 g'(w_5 x_1 + w_6 x_2) x_1$. If g is linear, the g' term is 1.
- Q3. B. Sigmoid saturates near 0 or 1, where its derivative is small.
- Q4. $\partial L / \partial W = X^T \Delta$, shape 20×5 .
- Q5. Too large LR can diverge or overshoot minima; poor validation may be data/model/loss issue rather than LR. Safer: LR range test, inspect train loss, overfit a tiny batch, check normalization/labels, monitor gradients.
- Q6. False. Adam often optimizes quickly, but generalization depends on data/model/regularization; SGD can generalize as well or better in some settings.
- Q7. $p_1 = e^2/(e^2 + 2e) = e/(e + 2)$ and $p_2 = p_3 = e/(e^2 + 2e) = 1/(e + 2)$. Gradient $p = (1, 0, 0)$.
- Q8. Residual blocks learn $F(x)$ with output $x + F(x)$, so identity mapping is easy. Gradients can pass through the skip path, reducing degradation/vanishing and allowing blocks to refine rather than completely transform features.
- Q9. Xavier keeps variance stable for roughly symmetric activations such as tanh; variance about $2/(n_{in} + n_{out})$. He uses about $2/n_{in}$ for ReLU-like activations that zero many negative values.
- Q10. Check input NaN/inf and normalization; reduce LR; inspect loss-output pairing; check log of zero/division; clip gradients; run full precision; overfit tiny batch; verify labels and class indices; print gradient norms and parameter ranges.

W3 Regularisation in Deep Learning

- Q1. C. Dropout masks activations during training and roughly behaves like averaging subnetworks with shared parameters.
- Q2. Examples: augmentation improves invariance but can change labels; weight decay shrinks parameters but can underfit; dropout reduces co-adaptation but slows training; early stopping is cheap but validation-noise sensitive; more data helps but costs.
- Q3. False. BN normalizes activations using batch statistics; dropout randomly masks activations. They have different mechanisms and evaluation behavior.
- Q4. Total gradient $0.4 + 2\lambda w = 0.4 + 0.3 = 0.7$. New $w = 3 - 0.1(0.7) = 2.93$.
- Q5. It teaches invariances and increases effective data diversity. It hurts when transformations change labels, distort medically/semantically important signals, or create train-test mismatch.
- Q6. A. BN has trainable scale and shift parameters.
- Q7. Likely overfitting. Save the best validation checkpoint, stop if validation metric fails to improve by a small tolerance for a patience window, and report final held-out test only after selection.



- Q8. L_1 has a constant-magnitude subgradient away from zero and encourages exact zeros/sparsity. L_2 smoothly shrinks but rarely makes them exactly zero.
- Q9. Tiny batch BN instability: layer/group norm or larger batch. Overconfidence: calibration/label smoothing. Model augmentation: WD/dropout/early stop. Exploding gradients: clipping/smaller LR/init.
- Q10. Using test for tuning leaks evaluation information and biases performance. Use cross-validation, nested validation, or split train into train/validation; reserve test for final estimate only.

W4 Convolutional Neural Networks

- Q1. Spatial size $\lfloor (15 - 3)/2 \rfloor + 1 = 7$. Output neurons $5 \times 7 \times 7 = 245$.
- Q2. CNN convolution uses fixed local shared kernels and strong translation/locality bias. Self-attention uses input-dependent pairwise token weights and can model global interactions in one layer, at higher token-quadratic cost.
- Q3. Spatial size $\lfloor (224 + 6 - 7)/2 \rfloor + 1 = 112$. Params $64(3 \cdot 7 \cdot 7 + 1) = 64(148) = 9472$.
- Q4. B.
- Q5. It improves gradient flow and lets blocks learn residual refinements. Projection is needed when spatial resolution or channel count changes so dimensions match for addition.
- Q6. False. A large filter is still a fixed learned local operation; attention is input-dependent and can connect arbitrary token pairs.
- Q7. VGG: stacked small convs, classification baseline. Inception: multi-scale branches, efficient classification. ResNet: residual learning, deep backbones. U-Net: encoder-decoder with skips, segmentation.
- Q8. Possible crop/position bias, insufficient translation augmentation, pooling/stride too aggressive, training distribution mismatch. Fix with random crop/translation, more diverse data, fully convolutional design, evaluation on shifted stress tests.
- Q9. IoU is intersection area divided by union area of predicted and true boxes. Detections count as correct only if class matches and IoU exceeds threshold; mAP averages precision-recall over thresholds/classes.
- Q10. Conv3->64 keeps 224×224 , ReLU; Conv64->64 keeps 224×224 , ReLU; maxpool stride 2 gives $(B, 64, 112, 112)$. Define layers in `__init__`, apply in forward.

W5 Graph Neural Networks

- Q1. CNNs exploit regular grids with local shared filters and are efficient for natural images. GCNs operate on nodes/edges and help when relationships are irregular: superpixels, object graphs, skeletons, scene relations, medical/anatomical structures, or long-range region dependencies.
- Q2. $H' = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H W)$ with $\tilde{A} = A + I$. Output shape $n \times k$.
- Q3. B. Graph readout should not change when nodes are re-ordered.
- Q4. Oversmoothing occurs when repeated aggregation makes node embeddings too similar. Mitigate with residual/skip connections, normalization, shallow depth, jumping knowledge, APPNP, or graph rewiring.
- Q5. False. Self-loops preserve and transform the node's own features during aggregation.
- Q6. Likely distribution shift and graph construction/degree bias. Evaluate with graph-level held-out splits, degree-stratified metrics, normalization, augmentation/rewiring, and report performance across graph sizes/degrees.
- Q7. GCN: normalized sum, simple spectral-style propagation. GraphSAGE: sampled neighbourhoods, inductive scalability. GAT: learned attention over neighbours, adaptive relations. GIN: sum+MLP, strong discriminative power.
- Q8. Message passing follows supplied edges. If edges connect irrelevant nodes or omit important relations, the model aggregates misleading information or cannot access needed context.
- Q9. About $50,000 \times 64 = 3.2$ million edge-hidden operations, ignoring constants and node transforms.
- Q10. Nodes are body joints, edges are bones and possibly temporal edges, features are joint coordinates/velocities/confidence, output action class. Failure modes: noisy pose estimation, missing joints, viewpoint shift, or inadequate temporal modelling.

W6 Consolidation: Gradients, Shapes, and Exam Reasoning

- Q1. Parameters: input-hidden weights $9H$, hidden biases H , hidden-output weights $5H$. Total $15H = 300$, so $H = 20$.
- Q2. Draw computation graph, assign forward variables, write local derivatives, multiply along paths, sum multiple paths, check shape/sign, then map derivative to update rule.
- Q3. $(AB)C$: $mnp + mpq$. $A(BC)$: $npq + mnq$. Choose the smaller depending on dimensions.
- Q4. B.
- Q5. False. It changes behavior of modules such as dropout and batch normalization; it does not update learned weights.



- Q6. Use mechanism, inductive bias, compute/memory, data requirements, strengths, failure modes, and concrete comparisons. Comparisons must explain why differences matter, not merely define names.
- Q7. $800/40 = 20$ updates per epoch, so 240 updates total.
- Q8. A sufficiently expressive model should drive training loss near zero on a tiny batch. Failure suggests bugs in data pipeline, gradient flow, LR, or architecture.
- Q9. No learning: check LR/loss/labels/gradients. Train good val bad: overfit/data shift. Both bad: underfit/bug/insufficient training. Noisy val: small validation set, seed variance, unstable metric.
- Q10. Define threat model; state method mechanism; give at least two advantages; give at least two limitations; mention compute/accuracy trade-off; state when it fails or needs adaptive evaluation.

W7 RNNs, LSTMs, and Transformers

- Q1. Gradients contain products of recurrent Jacobians across many time steps. If singular values are repeatedly below 1, gradients vanish; if above 1, they explode. Truncation, gating, residual paths, normalization, and clipping help.
- Q2. Detaching prevents the graph from spanning previous batches and consuming memory or backpropagating through unrelated history. `detach()` returns a detached tensor; `detach_()` detaches in place.
- Q3. $30 \cdot 50 + 50 \cdot 50 + 50 + 50 \cdot 10 + 10 = 1500 + 2500 + 50 + 500 + 10 = 4560$.
- Q4. A.
- Q5. $\text{softmax}(QK^T / \sqrt{d_k})V$. Scaling prevents dot products from growing too large with dimension, avoiding overly saturated softmax.
- Q6. False. Self-attention without positional information is permutation-equivariant and does not inherently know order.
- Q7. LSTM processes sequentially with gated state and useful recurrence bias; hard to parallelize but memory can be constant per step. Transformers train in parallel and model global pairwise relations, but full attention is $O(T^2)$ memory/time and needs position encoding.
- Q8. Full: $1024^2 = 1,048,576$ scores. Local: about $1024 \cdot 32 = 32,768$ scores, ignoring boundary effects.
- Q9. Training conditions on true previous tokens, while inference conditions on model outputs; errors compound. Mitigate with scheduled sampling, sequence-level training, or robust decoding/evaluation.
- Q10. RNN: hidden recurrence, vanishing/exploding. GRU: compact gates, less expressive than LSTM sometimes. LSTM: cell and gates, higher cost. Transformer: attention/KV, quadratic attention and data hunger.

W8 Beyond Transformer Architectures

- Q1. Full attention scales quadratically with sequence length, making long-context training/inference expensive in memory and time.
- Q2. B.
- Q3. SSM can update a compact state with low latency and linear cost, suitable for streaming. Full attention can directly compare all tokens but needs growing memory/cache. SSM may lose exact pairwise recall; attention may be too expensive.
- Q4. False. Efficiency does not guarantee recall; compressed state may forget details, so benchmarks must test retrieval and reasoning.
- Q5. It stores previous keys/values so each new token need not recompute all previous layers. Limitation: memory grows with context and batch size.
- Q6. Attention: pairwise token weights, quadratic cost. Mamba/SSM: selective recurrent state, possible recall loss. RWKV: RNN-like inference, smaller ecosystem. KAN: learnable edge functions, immature scaling/tooling.
- Q7. It scales as T^2 , so doubling T increases matrix size by $4\times$.
- Q8. Same parameter/data/budget comparison; hardware/runtime; retrieval at different positions, lost-in-middle tests, reasoning not just copying, ablations, OOD tasks, and quality/safety evaluation.
- Q9. Exact pairwise interaction can directly condition on every pair of tokens. Compressed-state models summarize past into a state, improving efficiency but risking information loss.
- Q10. Use SSM/local recurrence for efficient long-range scanning and attention/retrieval blocks for precise evidence alignment. Hybrid design balances cost, streaming, and exact cross-token reasoning.

W9 Multimodal Foundation Models

- Q1. Tasks need inputs, constraints, outputs, priority, and success criteria. A strong prompt specifies role, context, task list, constraints, evidence sources, output schema, and ambiguity-handling rules. Specificity reduces underspecified or hallucinated responses.
- Q2. A VLM processes visual and text inputs. Cross-attention lets tokens from one modality query keys/values from another, enabling grounded fusion.



- Q3. $(224/16)^2 = 14^2 = 196$ patch tokens.
- Q4. B.
- Q5. False. It may ignore, misread, or overgeneralize visual evidence, so hallucination remains possible.
- Q6. Use expert-labeled held-out data, measure diagnostic sensitivity/specificity, grounding accuracy, hallucination, subgroup performance, human review workflow, and failure cases. Avoid using it as autonomous diagnosis without validation.
- Q7. Early fusion combines tokens early for rich interaction but costly. Late fusion combines separate embeddings efficiently but weak interaction. Interleaved fusion alternates modality-specific and cross-modal blocks for balance.
- Q8. Different modalities have different raw structures/distributions. Contrastive objectives map matched examples near each other in a shared embedding space while separating mismatches.
- Q9. Causes: low resolution, patch compression, weak spatial grounding, biased captions, prompt ambiguity. Fix/test with higher-res crops, detection/grounding modules, counting benchmarks, chain-of-evidence prompts, and error analysis.
- Q10. Video adds temporal dimension: many frames or tubelets create many more tokens, requiring temporal modelling and more memory/compute.

W10 Deep Reinforcement Learning

- Q1. S states, A actions, P transition dynamics, R reward, γ discount factor controlling future reward value.
- Q2. $1 + 0.5(2) + 0.5^2(4) = 1 + 1 + 1 = 3$.
- Q3. A.
- Q4. True. The policy changes the state distribution and future rewards.
- Q5. The agent must exploit known good actions while exploring uncertain actions for better future return. Strategies include epsilon-greedy, entropy bonus, curiosity, or Thompson sampling.
- Q6. The robot may move fast while ignoring dirt or causing damage. Include cleanliness, safety penalties, energy/collision constraints, task completion, and human feedback; evaluate reward hacking scenarios.
- Q7. $y = 2 + 0.9 \cdot 3 = 4.7$.
- Q8. DQN: TD target, discrete/action instability. Policy gradient: returns/advantage, high variance. Actor-critic: critic estimates advantage, bias/instability. DPO/RLHF-style preference optimisation: preference pairs/reference or reward model, preference bias and limited coverage.
- Q9. Subtracting a state-dependent baseline from returns reduces variability of the learning signal. Since the expected score function times a baseline independent of action is zero, the gradient remains unbiased.
- Q10. It learns about a target policy from data generated by another behavior policy. It improves reuse/sample efficiency but can be unstable or biased without correction and coverage.

W11 Self-Supervised Representation Learning

- Q1. It encourages hidden features to be locally insensitive to small input perturbations, learning stable manifolds rather than memorizing noise.
- Q2. A.
- Q3. Linear probing freezes the backbone and tests separability of learned features. Fine-tuning updates all or most weights and tests adaptability to downstream task.
- Q4. There is 1 positive view from the same image. Total views 256; excluding itself and its positive leaves 254 potential negatives.
- Q5. False. Different samples can share semantic class; treating them as negatives can hurt representation learning.
- Q6. High masking forces semantic reconstruction rather than copying local texture. Encoding only visible patches reduces encoder compute, while a lightweight decoder reconstructs pixels.
- Q7. AE: reconstruct input, identity copying/blur. Contrastive: match augmentations, false negatives/collapse. MAE: reconstruct masked patches, pixel bias. CLIP: image-text alignment, caption noise/modality bias.
- Q8. Collapse means embeddings become identical or non-informative. Avoid with negatives, stop-gradient, predictor asymmetry, variance/covariance regularization, or EMA teacher.
- Q9. Pretrain on unlabeled images with MAE/contrastive augmentations relevant to products; evaluate via linear probe; fine-tune on labeled defects with class imbalance handling; validate on held-out product batches and inspect OOD failures.
- Q10. Positive pairs should preserve semantic identity. In some domains flips, crops, or color shifts can remove or alter label-relevant information.



W12 Deep Generative Models

- Q1. B. The optimal discriminator cannot distinguish real from generated and outputs 0.5.
- Q2. Diffusion is more stable, covers modes better, supports strong conditioning, and scales predictably. Limited by slow sampling, compute cost, physical/spatial inconsistency, object permanence, and audio/temporal alignment.
- Q3. $\min_G \max_D E_{x \sim p_d} \log D(x) + E_z \log(1 - D(G(z)))$. G generates samples to fool D ; D distinguishes real from generated.
- Q4. $\sqrt{0.81} = 0.9$ and $\sqrt{0.19} \approx 0.436$.
- Q5. A.
- Q6. True. The generator may produce limited sample diversity that fools the discriminator.
- Q7. GAN: sharp but unstable/mode collapse. VAE: latent likelihood-style training but blurry. Diffusion: high quality/diversity but slow. Flow matching: continuous transport, promising but newer.
- Q8. Use paired video-audio-text data; encode audio as spectrogram or audio tokens; condition/jointly generate video and audio latents with cross-modal attention; train sync/semantic losses; evaluate audio-video sync, perceptual quality, caption alignment, and safety.
- Q9. It combines conditional and unconditional scores during sampling. Higher guidance improves condition adherence but can reduce diversity and create artifacts.
- Q10. Small perturbations can move inputs across high-dimensional decision boundaries and produce large logit changes. Defences include adversarial training, randomized smoothing/certification, robust preprocessing, detection with abstention, and adaptive robust evaluation.